

The Decision Matrix

Which LLM on which hardware

AND WHEN "SOMEONE ELSE'S COMPUTER" IS THE HONEST ANSWER

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Local boxes, unified memory, consumer and workstation GPUs, H100 cloud, and API. One framework to pick between them.

Start with the decision tree

Do not start with GPUs.

Ask these first:

Fast question	It decides
data cannot leave?	local, EU sovereign cloud, or TEE API
frontier quality?	API first, hardware second
one active user?	local workstation can win
10+ active users?	serving stack before card name
less than 20 h/week?	rent or API
no operator?	managed cloud

Question 1: data

Can prompts, files, and logs leave the EU?

- no: local, EU sovereign cloud, or TEE API
- yes: API and GPU cloud stay on the table
- mixed: route by data class

TEE examples: NEAR AI, Venice.ai.

The cheapest question. It rules out half the options in one minute.

Question 2: quality

Is a local open model good enough?

- yes: optimize for fit and cost
- no: use API for the hard parts
- unknown: run a 20-task bake-off

Prototype your real workflow on a hosted API first. It is the cheapest way to learn if the quality is good enough, before you buy a box.

Question 3: users

How many people are generating at once?

Active users	Design answer
1	local box is fine
2 to 5	high-end local or rented GPU
10 to 50	vLLM, TensorRT-LLM, H100-class
50+	platform, not a machine

Question 4: model size

Pick the smallest model that does the job.

Model class	Practical floor
7B to 14B Q4	12GB VRAM or 32GB unified
30B to 32B Q4	24GB VRAM
70B Q4	64GB+ unified or 96GB VRAM
120B Q4	96GB VRAM or multi-GPU
MoE giants	total params still need memory

It has to fit twice: weights now, KV cache later.

Question 5: budget

Honest targets at street prices, June 2026:

SINGLE USER

under EUR 1k: 7B to 14B helper

EUR 2k to 4k: serious single-user box

EUR 4k to 6k: 5090, Mac Studio, DGX Spark class

BEYOND ONE BOX

EUR 13k+: RTX PRO 6000 workstation

EUR 50k+: team serving node

Budget EUR 4k, requirement 20 users on 70B? The requirement is the problem.

Entry options

Cheap does not mean bad. Use these for learning, private helpers, small tasks.

RTX 3060 12GB

Around EUR 340 new. Solid 7B to 8B floor.

MAC MINI M4

From around EUR 630. Quiet, efficient, memory is the limit.

MAC MINI M4 PRO 64GB

Around EUR 2.7k to 3.0k. More room for local context.

USED ENTERPRISE TRICKS

Fun, but fragile. Adapters, fans, driver patience.

Speed reality

Planning numbers, not lab promises:

Model class	What feels realistic
7B/8B Q4	3060 about 40 to 75, 5090 about 290 to 300
14B Q4	entry 35 to 60, fast CUDA 100+
30B/32B Q4	24GB to 32GB CUDA, about 25 to 80
70B+	memory first: M3 Ultra 70B Q4 about 16.5

Short context. Long context drags every number down.

At EUR 4.0k, 5090 beats a new EUR 3.5k 4090.

Unified memory

Sometimes fit beats speed. These are the "model fits" machines.

STRIX HALO 128GB

Framework Desktop near EUR 2.3k, GMKtec EVO-X2 near EUR 3.7k. Ryzen AI Max+ 395, 112GB to the GPU.

MAC STUDIO M3 ULTRA 96GB

Around EUR 4.3k+. 819 GB/s memory bandwidth.

DGX SPARK

Around EUR 4.8k to 5.5k. CUDA desk appliance.

ASUS ASCENT GX10

Around EUR 3.6k to 4.3k. Cheaper GB10 variant.

Workstation breakpoint

RTX PRO 6000 Blackwell:

- 96GB GDDR7
- around EUR 12.8k to 13.1k
- 70B becomes comfortable
- 120B Q4 becomes plausible
- single-card simplicity

You pay to make multi-GPU pain obsolete.

Above one big card

For serious local and on-prem work:

Need	Pareto answer
small team node	4x RTX PRO 6000 Max-Q
deskside supercomputer	DGX Station GB300
many users, steady load	DGX/HGX B200
rack-scale frontier	GB300 NVL72 class

Past one GPU, the hard parts are cooling, interconnect, serving software, and operations.

How to run it

One user, one box. Start simple, then specialize.

EASY ON-RAMPS

Ollama: one command, CLI first.

LM Studio: GUI, GGUF or MLX, swap models fast.

For 7B to 70B chat and small task complexity.

POWER AND NEAR-FRONTIER

oMLX: MLX server on Apple Silicon, serves several models from one endpoint, drives Claude Code and Cursor.

ds4: a DeepSeek V4 Flash engine, 1M context, 2-bit fits 128GB.

Same model, different runtime, different limits. The runtime decides context, speed, and what fits.

Team serving

For teams, the card is not the product. You need:

- continuous batching
- chunked prefill
- KV cache control
- admission limits
- logs and billing

A one-lane runtime on a beautiful machine is still one lane.

Cloud math in Europe

EU H100 pricing anchors:

- Scaleway H100: from EUR 2.73/h
- OVHcloud H100: from EUR 2.80/h
- Nebius H100: about EUR 3.4/h

20 hours a week is about EUR 243 a month. 24/7 is about EUR 2,016. Rent until the box stops sitting idle.

Own vs rent

OWN WHEN

- load is steady
- data is sensitive
- latency matters
- someone can run it

RENT OR API WHEN

- load is bursty
- frontier quality matters
- ops time is scarce
- the box would sit idle

The decision in one page

1

Data cannot leave?

Local, EU sovereign cloud, or TEE API.

2

Frontier quality?

API or managed model.

3

One user?

Buy for fit and comfort.

4

A team?

Buy a serving stack, not a GPU.

5

Low utilization?

Rent.

6

Steady, private, staffed?

Own.



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